

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460141.9009 +/- 0.0077 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.06
Transit depth (Rp/Rs)^2	1.87 +/- 1.65 [%]
Orbital Inclination (inc)	83.8 +/- 2.78
Airmass coefficient 1 (a1)	2.94 +/- 1.76
Airmass coefficient 2 (a2)	-0.56 +/- 0.5
Scatter in the residuals of the lightcurve fit is	65.46 %
Best Comparison Star	None
Optimal Aperture	1.82
Optimal Annulus	3.75
Transit Duration (day)	0.052 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.137 $\pm$ 0.06 vs 0.1646 (deviation 0.39 σ)
Duration fit vs published	0.052 d vs 0.0758 d (deviation 0.7 σ)
Mid-time O-C	19.0 min
Depth SNR	2.0
Residual scatter	65.46 %

Light curve

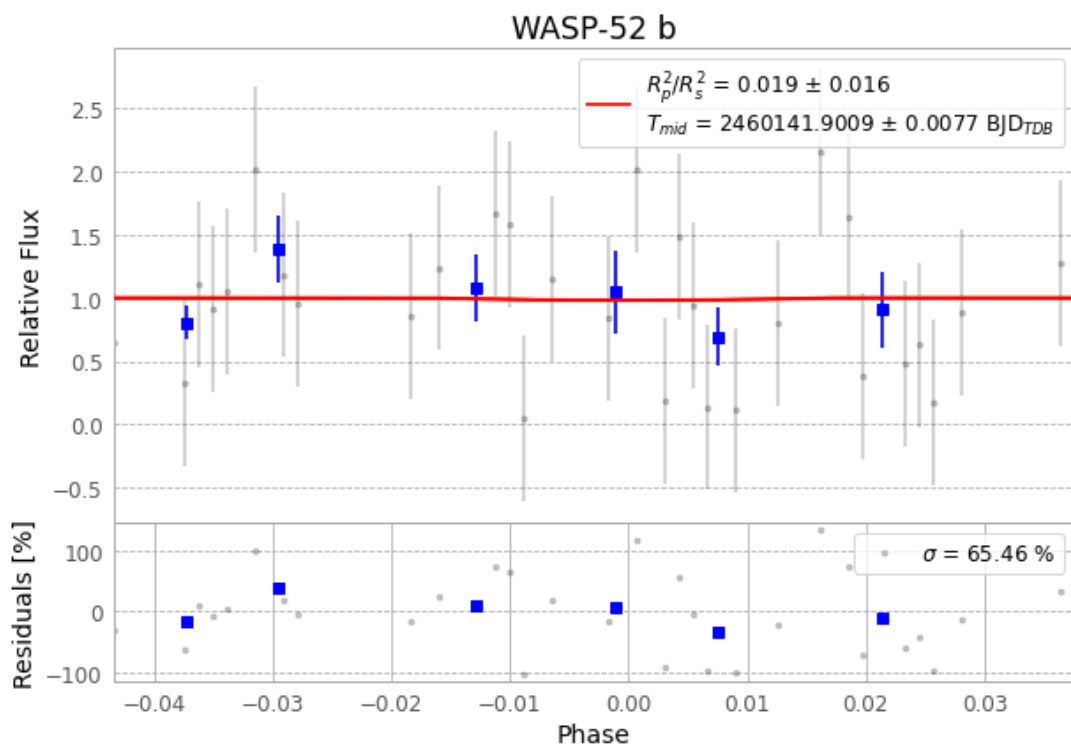


Figure 1 — FinalLightCurve_WASP-52 b_2023-07-16.png (SHA-256 6ec8ee02712586f1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [320, 151], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 140983ec0dfd2eff...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 72f1d3a

Data provenance

77 FITS frames (51 MB), WASP-52230716072115.FITS → WASP-52230716110915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-230716.log (35a1072d81b1...), FinalLightCurve_WASP-52 b_2023-07-16.png (6ec8ee027125...), FinalLightCurve_WASP-52 b_2023-07-16.csv (3b4867c4dfd7...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 04:38Z.